



Nanostructured Ag surface fabricated by femtosecond laser for surface-enhanced Raman scattering

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ABSTRACT

Femtosecond laser was employed to fabricate nanostructured Ag surface for surface-enhanced Raman scattering (SERS) application. The prepared nanostructured Ag surface was characterized by field emission scanning electron microscopy (FESEM), atomic force microscopy (AFM), X-ray diffraction (XRD), and X-ray photoelectron spectroscopy (XPS). The FESEM images demonstrate the formation of nanostructure-covered femtosecond laser-induced periodic surface structure, also termed as ripples, on the Ag surface. The AFM images indicate that the surface roughness of the produced nanostructured Ag substrate is larger than the untreated Ag substrate. The XRD and XPS of the nanostructured Ag surface fabricated by femtosecond laser show a face centered cubic phase of metallic Ag and no impurities of Ag oxide species. The application of the produced nanostructured Ag surface in SERS was investigated by using rhodamine 6G (R6G) as a reference chemical. The SERS intensity of R6G in aqueous solution at the prepared nanostructured Ag surface is 15 times greater than that of an untreated Ag substrate. The Raman intensities vary linearly with the concentrations of R6G in the range of 10^{-8} – 10^{-4} M. The present methodology demonstrates that the nanostructured Ag surface fabricated by femtosecond laser is potential for qualification and quantification of low concentration molecules.

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1. Introduction

Materials in nanometer scale may exhibit unique physical and chemical properties and have been explored for various applications such as surface-enhanced Raman scattering (SERS) [1–6]. Molecules adsorbed on nanostructured silver, gold, and copper surfaces are found to have an unusually large Raman cross-section under certain conditions. Therefore, the fabrication of controlled and reproducible metallic nanostructures where the target molecules are appropriately located for high Raman enhancement is of great interest. The SERS phenomenon arises from the electromagnetic contribution caused by the resonant excitation of surface plasmons of roughened metallic surface and the chemical contribution resulting from the charge transfer between molecule and substrate [7,8]. The SERS is ideally suited for label-free trace analysis such as the detection of pharmaceuticals, nerve stimulants, and environmental pollutants [9–11]. The preparation of nanostructured Ag surface has been a subject of intensive investigation because of its excellent SERS, which differentiates it from bulk materials. Methods for prepa-

ration of nanostructured Ag surface include Langmuir–Blodgett technique [12], thermal decomposition [13], photochemical deposition [14], nanoscale lithography [15], and electrochemical deposition [16–18]. Each of these methods has certain benefits associated it but they often require multiple steps and impurity might be involved during the preparation. Recently, femtosecond laser nanostructuring process has been demonstrated to be a promising technique for surface modification because of the low threshold for material removal from surfaces [19–21]. The generation of nanostructure-covered laser-induced periodic surface structures, also termed as ripples, on various materials upon laser irradiation at near-damage threshold fluence has been reported in the literature [22–24]. The period of laser-induced periodic surface structures is close to or less than the wavelength of the incident radiation and their orientation is perpendicular to the direction of the electric field [25]. The one-step femtosecond laser nanostructuring process for the fabrication of roughened Ag surface is considered as an effective procedure when compared with other methodologies. Here we demonstrate a facile route for the fabrication of Ag nanostructure-covered femtosecond laser-induced periodic surface structures for SERS chemical sensor application. Rhodamine 6G (R6G) is chosen as a probe molecule.

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2. Materials and methods

2.1. Reagents

R6G was obtained from Fluka. All solutions were prepared with demineralized and filtered water of resistivity of not less than 18 M Ω cm which was taken from a Milli-Q water purification system (Milli-Q, USA).

2.2. Apparatus

Raman spectra were recorded with a micro-Raman spectrometer (Jobin-Yvon Triax 550), equipped with a 25 mW He/Ne laser with an emission at 632.8 nm. The laser beam was focused at a nanostructured Ag substrate in a liquid cell containing 35 mL aqueous solution. Field emission scanning electron microscopy (FESEM) images were performed using an Ultra Plus (Zeiss, Germany). Atomic force microscopy (AFM) images were obtained using a SPA-400 (Seiko, Japan) multiple function units together with SPI-3800 N control station in tapping mode in the air. X-ray diffraction (XRD) patterns were recorded using a D8 Discover (Bruker) X-ray diffractometer with Cu K α radiation. X-ray photoelectron spectroscopy (XPS) measurements were obtained on a Phi 5000 VersaProbe Scanning ESCA Microprobe (Ulvac-Phi) instrument using Al K α radiation.

2.3. Fabrication of nanostructured Ag surface by femtosecond laser

Ag substrates were polished with 0.3 and 0.05 μ m aluminum oxide slurries and were rinsed with deionized water before surface modification. Linearly polarized laser pulses with a central wavelength of 800 nm emitted by a commercial Ti:sapphire femtosecond laser system (Spitfire, Spectra-Physics) were used for the laser scanning experiments in air at atmospheric pressure. Laser pulses with a duration of 120 fs were used for nanostructuring process with repetition rate of 1 kHz. The laser beam was loosely focused onto an Ag substrate mounted on an X-Y positioning stage at normal incidence to achieve a fluence of 0.34 J cm $^{-2}$. The nanostructured Ag substrates were produced by scanning (0.15 mm s $^{-1}$) a femtosecond laser beam across the Ag substrates with an area of 300 $\times$ 100 μ m 2 . After the laser irradiation experiments, the Ag substrates were cleaned using an ultrasonic water bath for 5 min.

3. Results and discussion

The morphological change of the femtosecond laser-irradiated Ag surface was investigated by FESEM and AFM. The FESEM images of the surface of Ag substrate before and after irradiating with femtosecond laser are shown in Fig. 1a and b, respectively. It can be seen from Fig. 1b that the femtosecond laser-induced periodic surface structures with spatial periods is less than the wavelength of the incident radiation and their orientation is perpendicular to the direction of incident laser beam. The inset of Fig. 1b demonstrates that the femtosecond laser-induced periodic surface structures are covered by random nanostructures which include nanocavities and nanospheres. It is important to note that the surface roughness of the produced nanostructured Ag surface is crucial for SERS applications. AFM was used to measure the geometry of the produced nanostructured Ag surface with a high lateral and vertical accuracy. The AFM images of the surface of Ag substrate before and after treatment using a femtosecond laser nanostructuring process are shown in Fig. 2a and b, respectively. The period of the produced nanostructured Ag surface is measured to be 560 nm which is less than the laser wavelength (see Fig. 2b). This is attributed to an increase of the real part of the effective index of the air–metal interface when surface roughness develops and affects the propagation

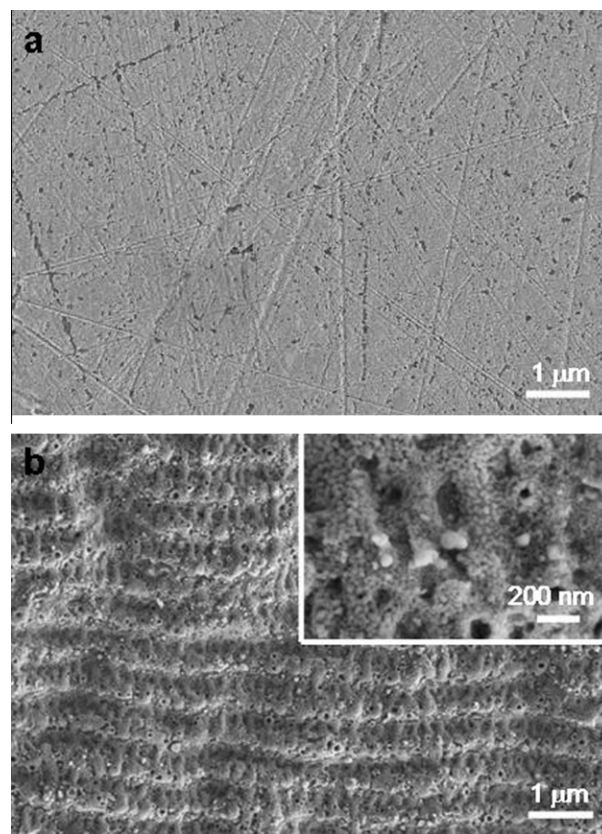


Fig. 1. FESEM images of the surface of Ag substrate before (a) and after (b) irradiating by femtosecond laser.

of surface plasmons [26]. The root-mean-square values are 2.9 and 43.8 nm for the surfaces of Ag substrate before and after treatment using a femtosecond laser nanostructuring process. From this data we can conclude that the surface roughness is larger for the nanostructured Ag surface fabricated by femtosecond laser. The nanostructured Ag surface prepared by this methodology might be suitable for SERS applications.

XRD and XPS were employed to confirm the crystal structure and chemical composition of metallic Ag after the scanning of femtosecond laser beam onto the surface of Ag substrate. XRD is a bulk analysis that reveals the crystal structure. The XRD pattern of Ag surface after treatment with femtosecond laser is shown in Fig. 3. As can be seen from Fig. 3, there are five peaks located at $2\theta = 38.2^\circ, 44.4^\circ, 64.5^\circ, 77.5^\circ$, and 81.6° , respectively. The five diffraction peaks can be indexed as (1 1 1), (2 0 0), (2 2 0), (3 1 1), and (2 2 2) planes of the face centered cubic Ag [27]. XPS was also performed for a more detailed characterization of the chemical bonding at the nanostructured Ag surface fabricated by femtosecond laser. The XPS core-level spectra for Ag 3d $_{5/2-3/2}$ on Ag surface after scanning with femtosecond laser is shown in Fig. 4. The XPS curve can be well fitted by two peaks centered at 368.2 and 374.2 eV which corresponds to Ag 3d $_{5/2}$ and 3d $_{3/2}$, respectively [27]. No apparent XPS peaks of AgO (367.4 eV) or Ag $_2$ O (367.8 eV) were observed at the produced nanostructured Ag surface, indicating that the Ag nanostructures induced by femtosecond laser are of metallic nature. This result is in good agreement with the XRD characterization. All the results validate the effectiveness of using femtosecond laser in producing nanostructured Ag surface.

To investigate the SERS behavior at the nanostructured Ag surface fabricated by femtosecond laser, the prepared nanostructured Ag substrate was placed on the bottom of a liquid cell with 35 mL

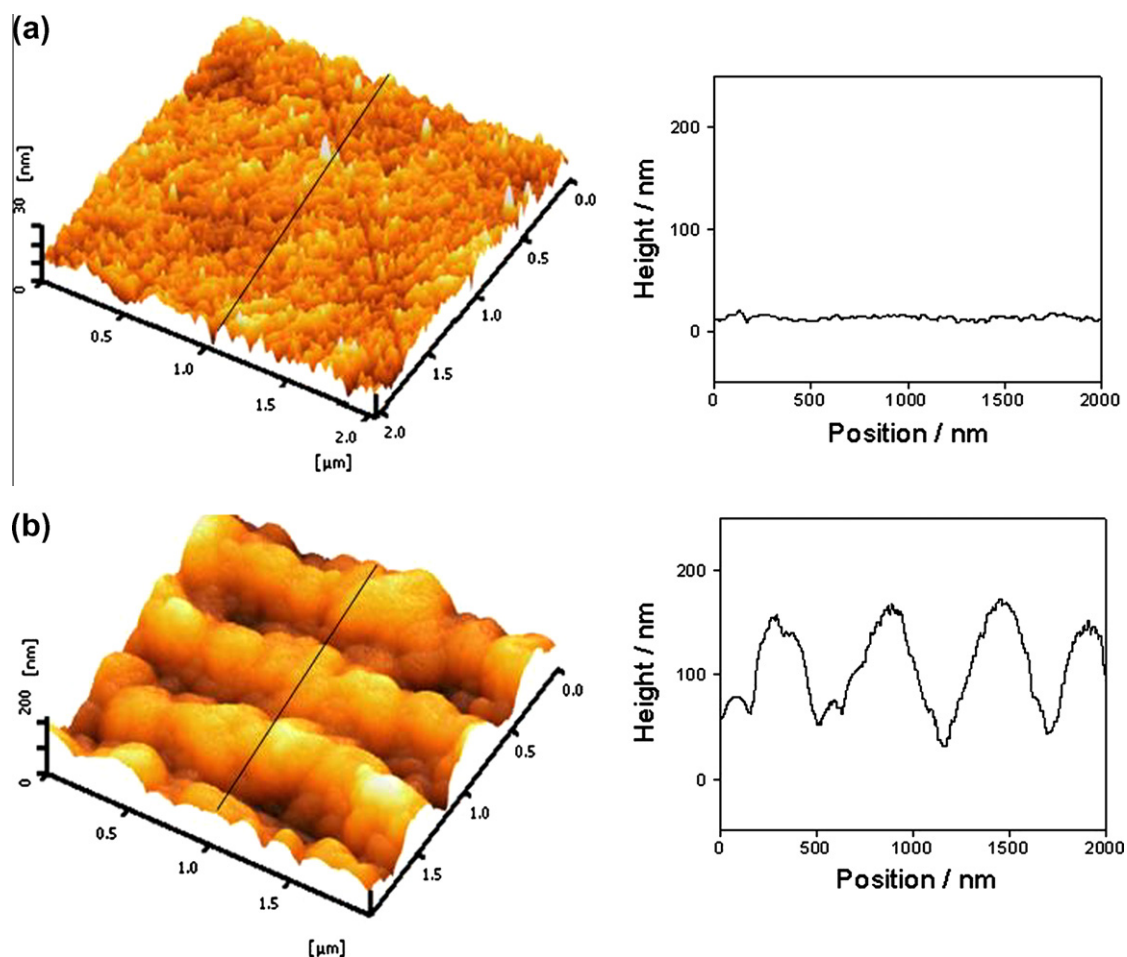


Fig. 2. AFM images of the surface of Ag substrate before (a) and after (b) irradiating by femtosecond laser.

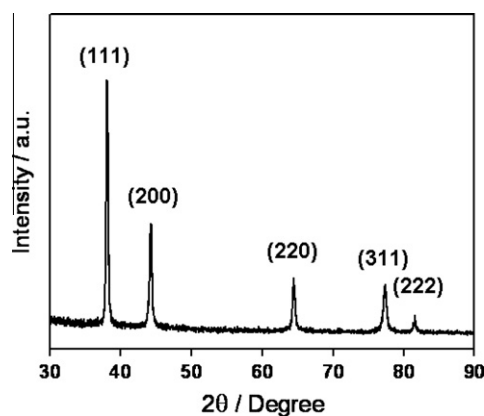


Fig. 3. XRD pattern of the nanostructured Ag surface fabricated by femtosecond laser.

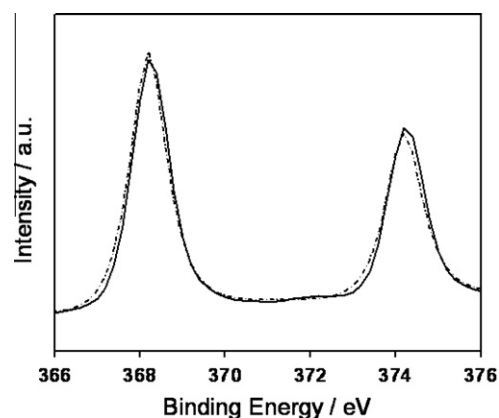


Fig. 4. XPS pattern of the nanostructured Ag surface fabricated by femtosecond laser.

aqueous solution containing 10^{-6} M of R6G. For comparison, control experiments at the untreated Ag substrate and silicon substrate were performed. The Raman spectra of 10^{-3} M R6G in aqueous solution at silicon substrate and 10^{-6} M R6G in aqueous solution at the polished Ag substrate and nanostructured Ag substrate fabricated by femtosecond laser are shown in Fig. 5a–c, respectively. No clear Raman signal of 10^{-3} M R6G was observed from silicon substrate. In contrast, SERS signal of 10^{-6} M R6G was obtained at the nanostructured Ag surface fabricated by femtosecond laser. Several strong peaks at 1186, 1310, 1362, 1509, and

1650 cm^{-1} are observed at the nanostructured Ag surface fabricated by femtosecond laser. The strong peaks at ca. 1362, 1509, and 1650 cm^{-1} are assigned to the stretching modes of aromatic C–C. The band at 1186 cm^{-1} is assigned to aromatic C–H bending. The characteristic Raman peaks in Fig. 5 were in agreement with other works [28,29,13]. The polished Ag surface gave a very small enhancement compared with the nanostructured Ag surface fabricated by femtosecond laser which was 15 times of magnitude more efficient. The enhancement factor at the nanostructured Ag surface was estimated to be 1.2×10^6 as compared to silicon substrate. The

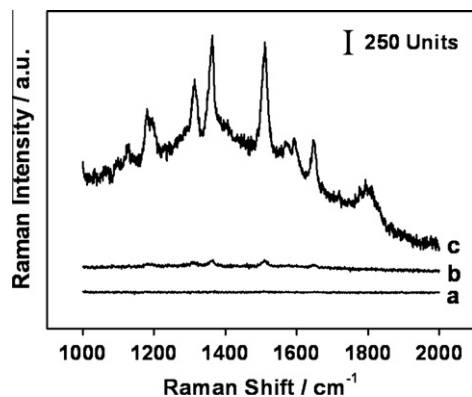


Fig. 5. Raman spectra of 10^{-3} M R6G in aqueous solution at silicon substrate (a) and 10^{-6} M R6G in aqueous solution on the surfaces of Ag substrate before (b) and after (c) irradiating by femtosecond laser. The spectra were recorded using a 25 mW He/Ne laser with an emission at 632.8 nm. The integration time was 1 s.

enhancement factor is in the similar range at SERS-active substrates prepared by femtosecond laser followed by Ag chemical plating on a silica glass substrate [30]. The results imply that the femtosecond laser is effective for the preparation of SERS-active surface. The detection sensitivity for SERS at the nanostructured Ag substrate is competitive as compared to SERS-active substrates prepared by femtosecond laser followed by Ag chemical plating on a silica glass substrate [30], femtosecond laser induced co-deposition of Ag nanoparticles and R6G on a glass substrate [31], and femtosecond laser direct writing on a Ag⁺-doped phosphate glass followed by Ag chemical plating [32]. To evaluate the potential applications of the produced nanostructured Ag surface for in situ determination of R6G in aqueous solution, different concentrations of R6G were added into the liquid cell. The Raman spectra obtained at the prepared nanostructured Ag surface to the additions of varying concentrations of R6G in aqueous solution are shown in Fig. 6. It can be seen that the Raman intensities increased with successive additions of R6G. The calibration curve of Raman peak intensity (at 1509 cm^{-1}) at the nanostructured Ag substrate fabricated by femtosecond laser to the additions of varying concen-

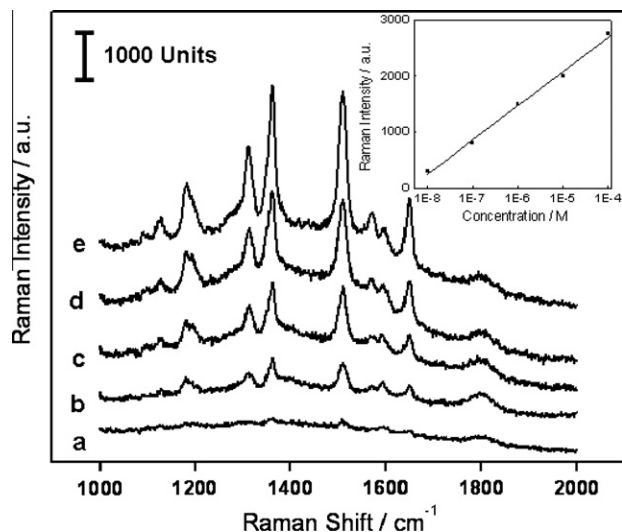


Fig. 6. Raman spectra obtained at the nanostructured Ag surface fabricated by femtosecond laser for successive additions of R6G in aqueous solution. R6G: 10^{-8} (a), 10^{-7} (b), 10^{-6} (c), 10^{-5} (d), and 10^{-4} (e) M. The inset shows the calibration curve of Raman peak intensity (at 1509 cm^{-1}) of R6G obtained at the nanostructured Ag substrate fabricated by femtosecond laser. The spectra were recorded using a 25 mW He/Ne laser with an emission at 632.8 nm. The integration time was 1 s.

trations of R6G in aqueous solution is shown in the inset in Fig. 6. The Raman intensities vary linearly with the concentration of R6G in the range of 10^{-8} – 10^{-4} M with a correlation factor of 0.997 at an integration time of 1 s. This suggests that this approach is effective and sensitive for the preparation of active SERS substrate.

4. Conclusions

We have demonstrated an effective one-step procedure for the preparation of SERS-active substrate by scanning a laser beam across an Ag surface. XRD and XPS characterization studies show that the Ag nanostructure-covered femtosecond laser-induced periodic surface structure is in metallic state without the introducing of other impurities of Ag oxide species. The in situ determination of R6G in aqueous solution can be achieved by using this methodology. The advantages of the preparation of nanostructured Ag surface by femtosecond laser over other forms of SERS-active substrates are no contamination, procedural simplicity, and capability to process complicated shapes. This method may be extended to prepare novel nanostructures of other noble metals for the potential applications on developing sensitive optical chemical sensors, nanoelectronics, and biomedical devices.

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